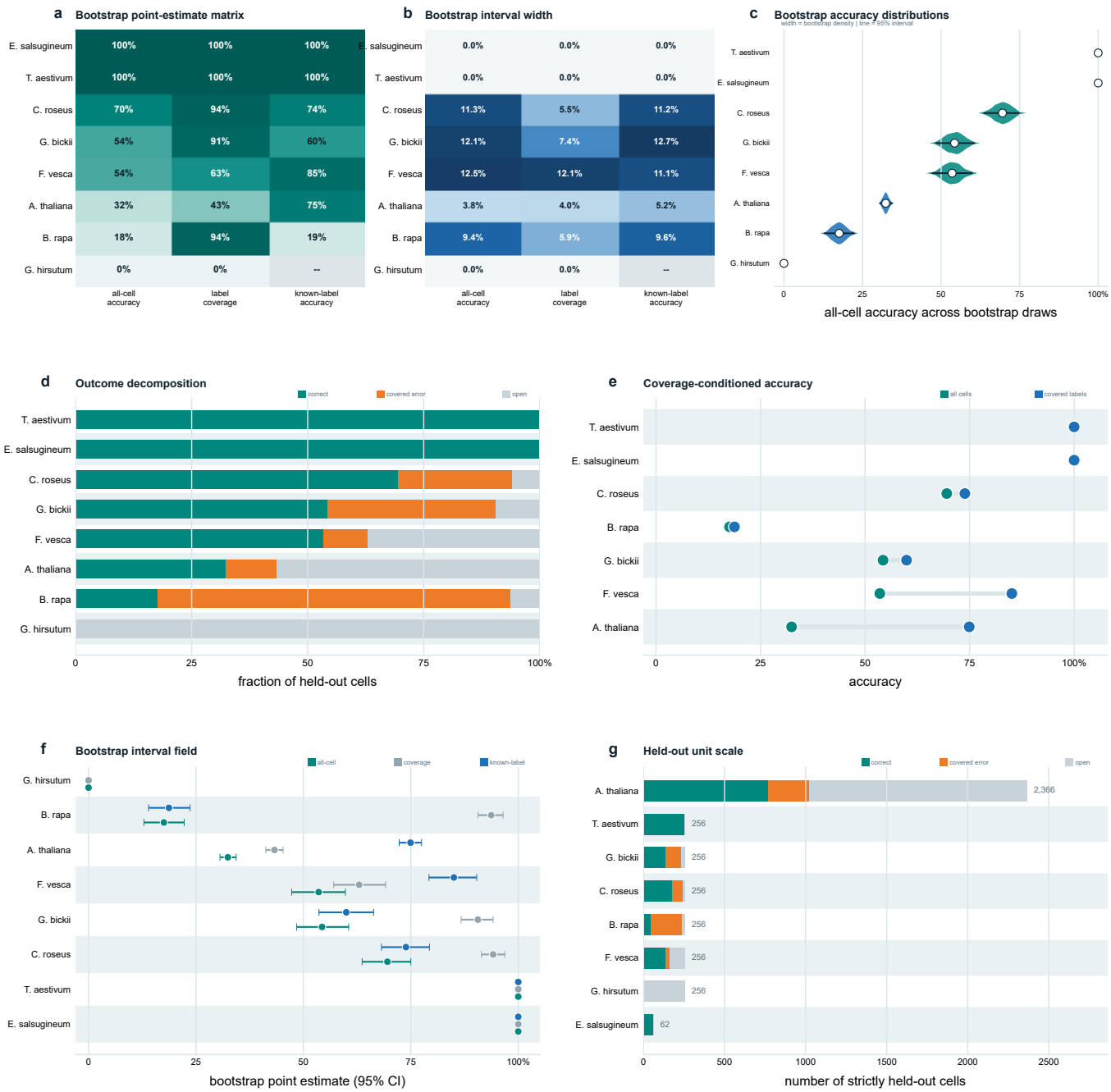
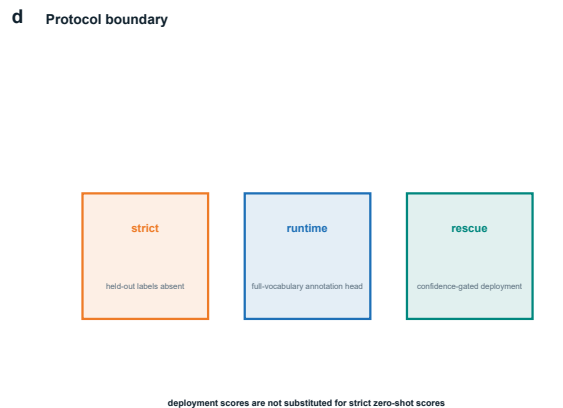
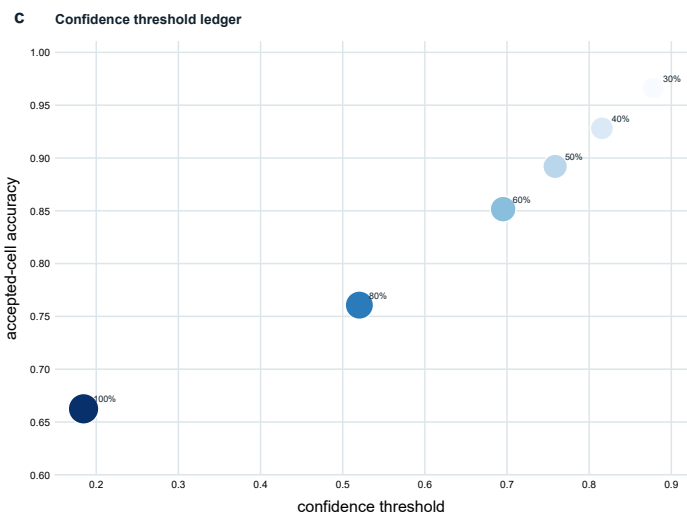
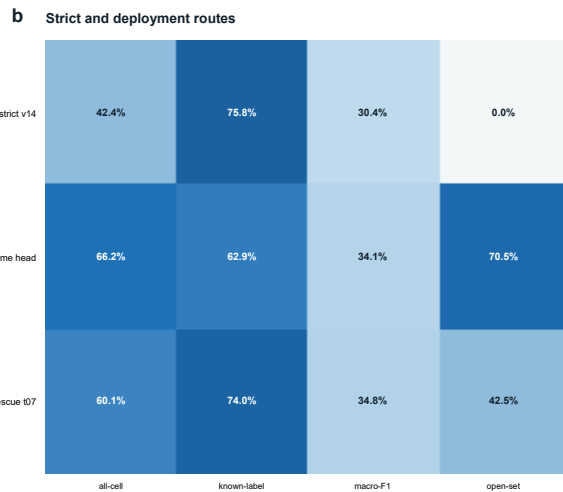
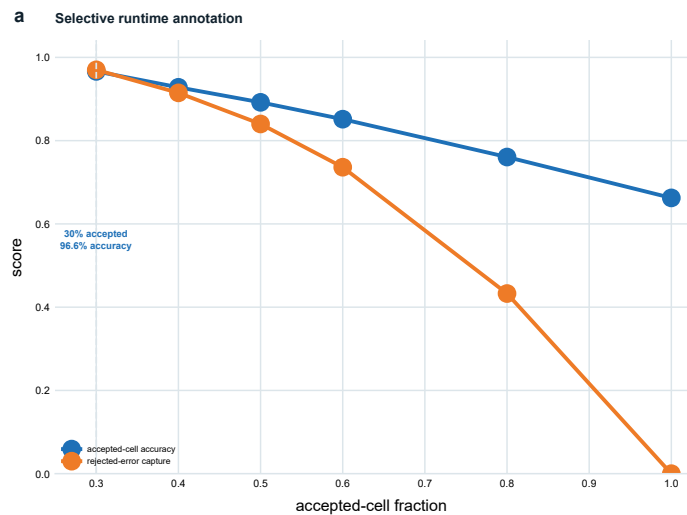


Strict leaf-species: uncertainty and label coverage



Runtime confidence and deployment boundary

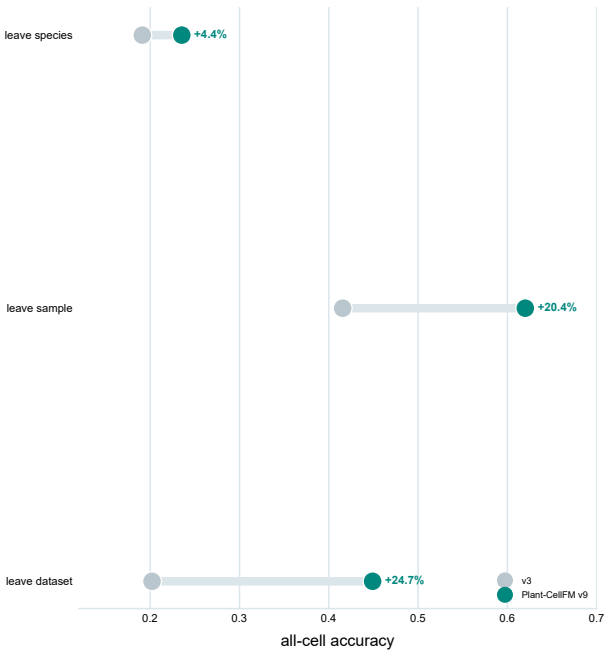


deployment scores are not substituted for strict zero-shot scores

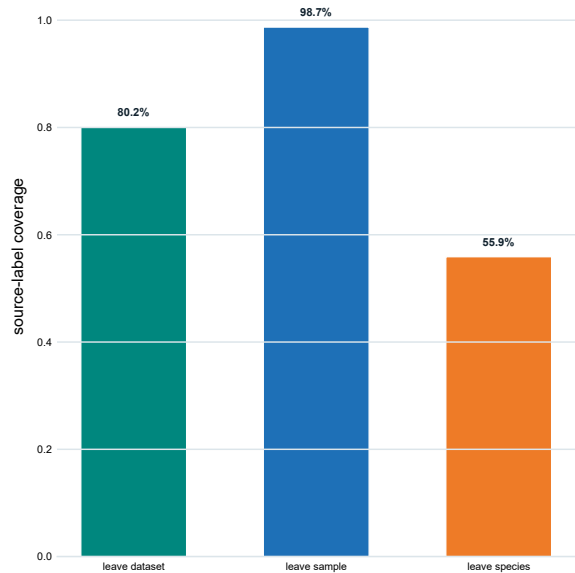
Matched internal checkpoint comparison

shared data, split and label contract; not a third-party ranking

a Matched frozen checkpoint gain



b Shared-protocol coverage



Sorghum root: exchanged sealed-library tests

same 27-state author label space

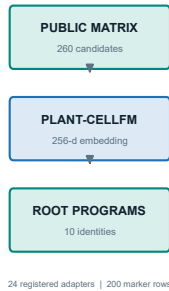


Sorghum root: state-resolved error atlas

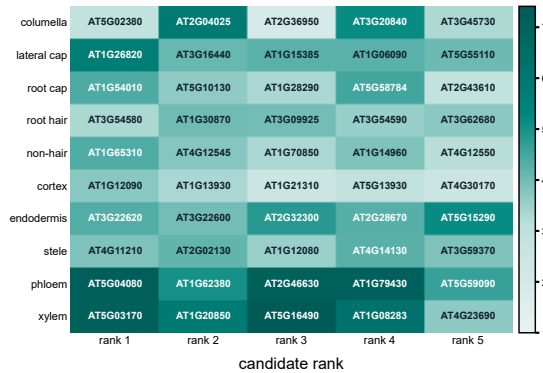
sealed OUGHW test | 4,150 cells | 27 author states



a Evidence route



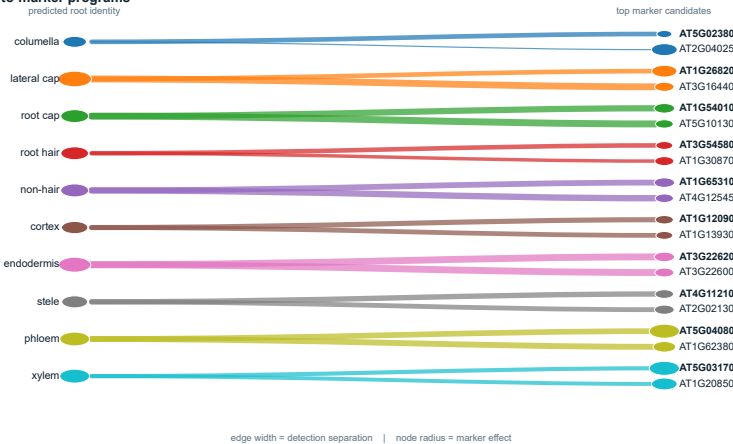
b Top marker programs



c Marker effect geometry



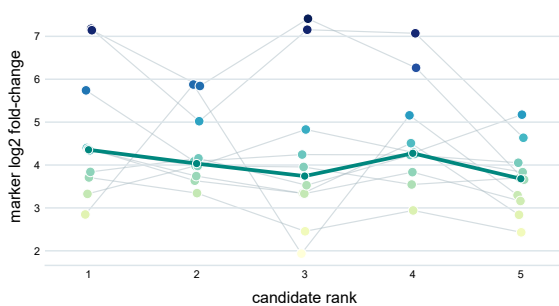
d Identity-to-marker programs



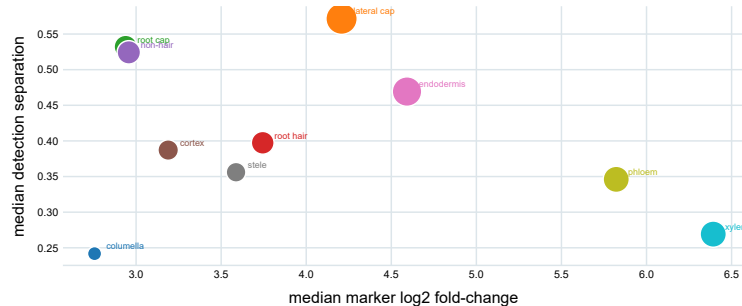
e Identity-level effect summary

columella	2.76	24%	0.61
lateral cap	4.21	57%	2.47
root cap	2.94	53%	1.40
root hair	3.74	40%	1.37
non-hair	2.96	52%	1.44
cortex	3.19	39%	1.13
endodermis	4.59	47%	2.21
stele	3.59	36%	1.06
phloem	5.82	35%	1.77
xylem	6.39	27%	1.77
	median log2FC	detection delta	marker score

f Rank-wise marker effect

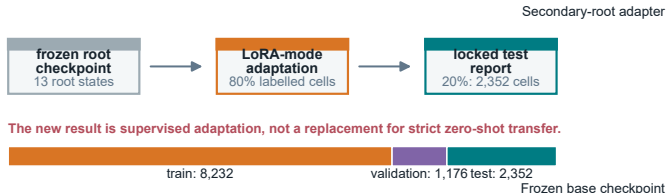


g Identity-level effect geometry



a A context-specific state layer augments the frozen checkpoint

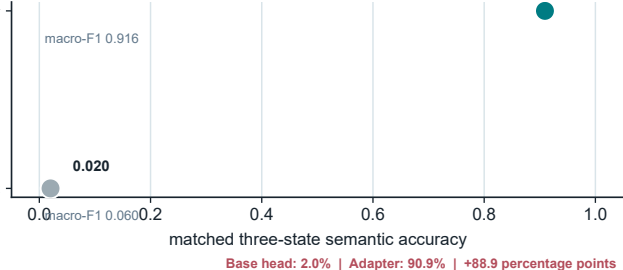
GSE270140/GSM8335426 author labels: 11,760 cells, 14 states, one secondary-root sample



The new result is supervised adaptation, not a replacement for strict zero-shot transfer.

b Labelled adaptation recovers held-out vascular states

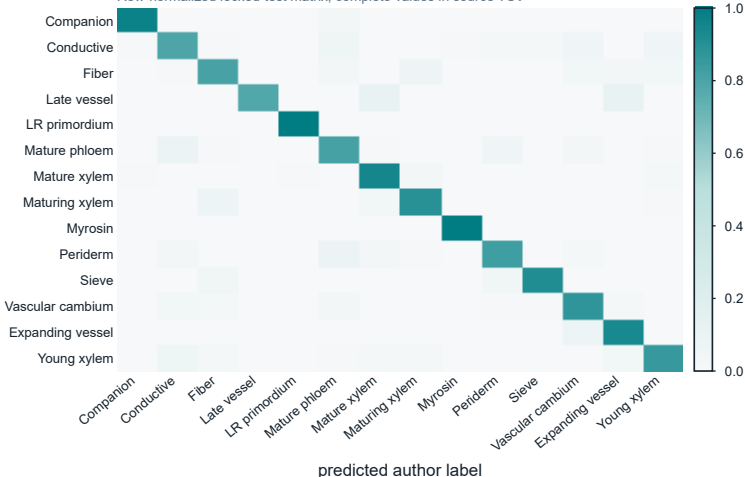
1,885 shared-state cells; fixed-bootstrap 95% intervals



Base head: 2.0% | Adapter: 90.9% | +88.9 percentage points

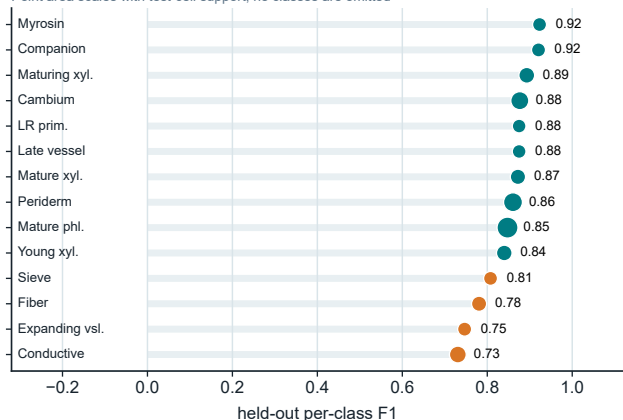
c All 14 secondary-root states resolve on the locked test

Row-normalized locked-test matrix; complete values in source TSV



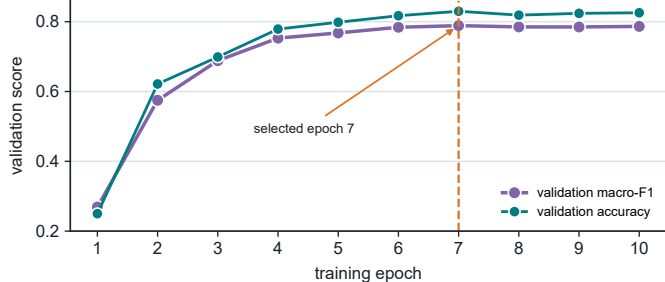
d Per-class F1 preserves rare-state visibility

Point area scales with test-cell support; no classes are omitted



e Validation plateaus before locked-test evaluation

Best checkpoint is selected exclusively by validation macro-F1



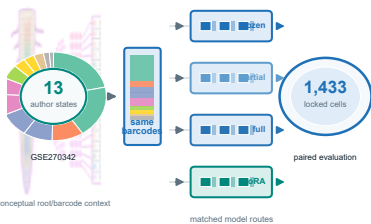
f Calibrated held-out results support review-aware use

Reported separately from strict transfer and from label-free external execution

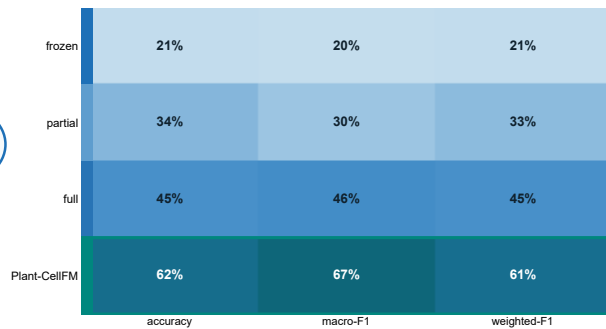


Test evaluator: 2,352 cells; original one-sample study design is retained in the record.

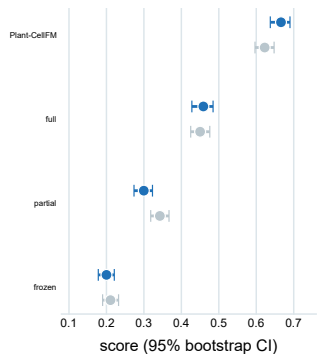
a Sealed study design



b Matched benchmark field



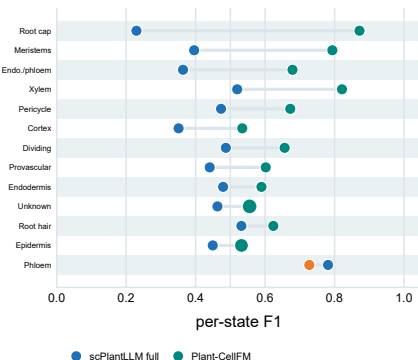
c Paired uncertainty



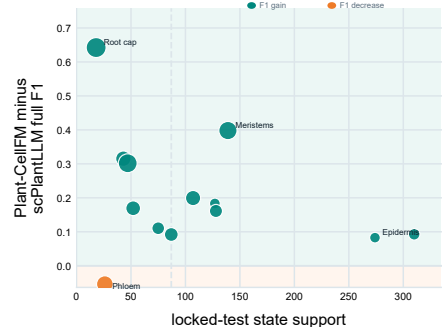
d State performance matrix

Root cap	12%	0%	23%	87%
Phloem	28%	82%	78%	73%
Xylem	33%	48%	82%	82%
Meristems	19%	28%	40%	79%
Endo./phloem	14%	7%	36%	68%
Pericycle	18%	27%	47%	67%
Dividing	12%	18%	49%	66%
Root hair	38%	43%	83%	82%
Provascular	12%	30%	44%	80%
Endodermis	19%	7%	48%	89%
Unknown	37%	49%	46%	56%
Cortex	12%	23%	35%	83%
Epidermis	11%	33%	45%	53%
	frozen	partial	full	Plant-CellFM

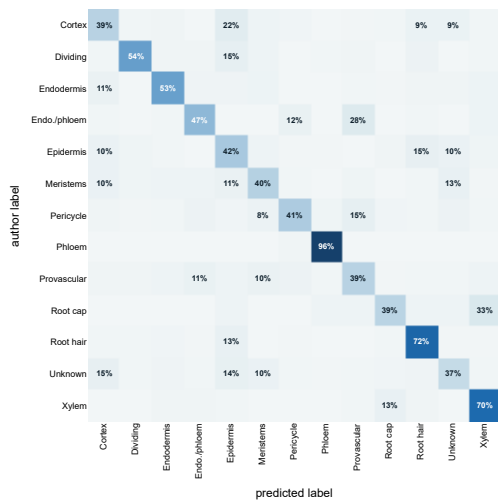
e State-level gain



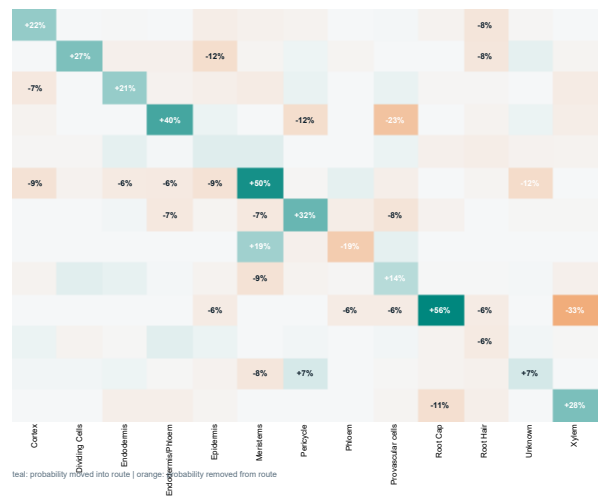
h Gain x state support



f scPlantLLM full: error topology

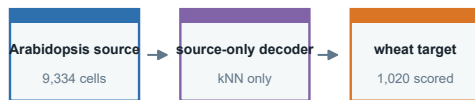


g Error-route change: Plant-CellFM - scPlantLLM



a Zero-target labels remain physically separated

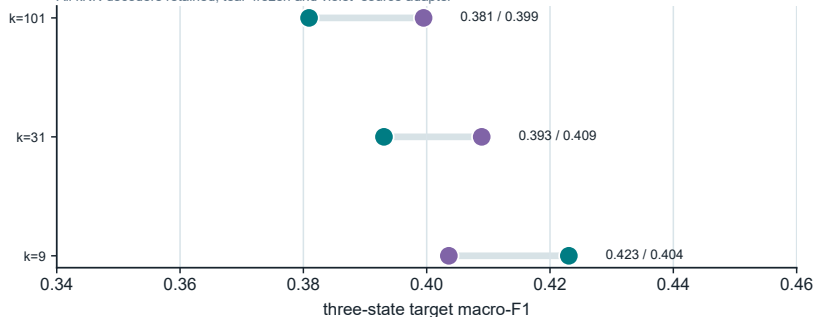
The source-only decoder is written before wheat labels are read for scoring



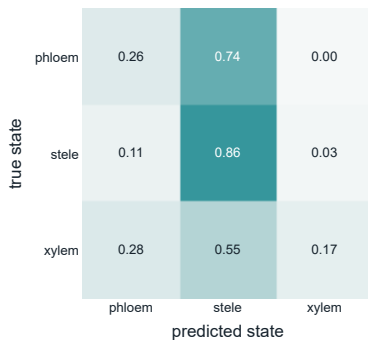
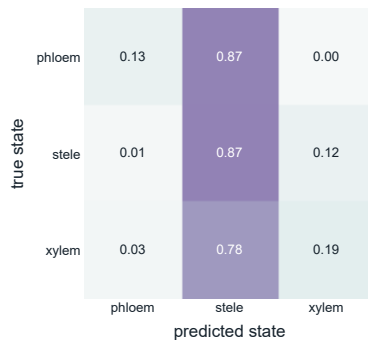
No wheat labels are used for fitting, calibration or decoder selection.

b Source adaptation does not improve zero-target macro-F1

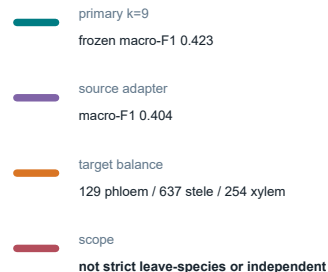
All kNN decoders retained; teal=frozen and violet=source adapter

**c****Cross-species vascular-state errors remain inspectable**

Row-normalized three-state confusion; all target classes are retained

Frozen root checkpoint (k=9)**GSE270140 source adapter (k=9)****d Why the negative result remains**

Retaining the failure prevents a selective cross-species narrative



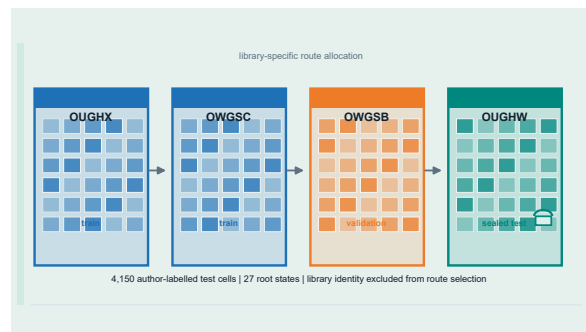
Independent evaluation contracts

target-label isolation | library separation | independent external cells | matched barcodes

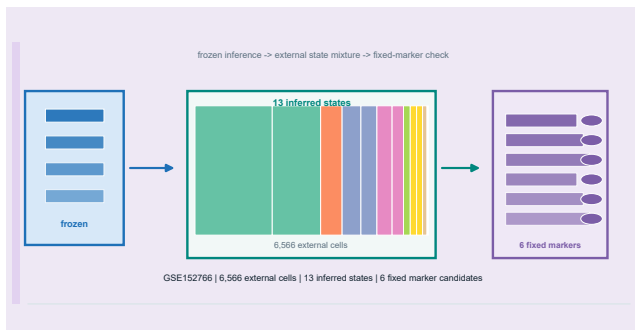
a Strict leave-species contract



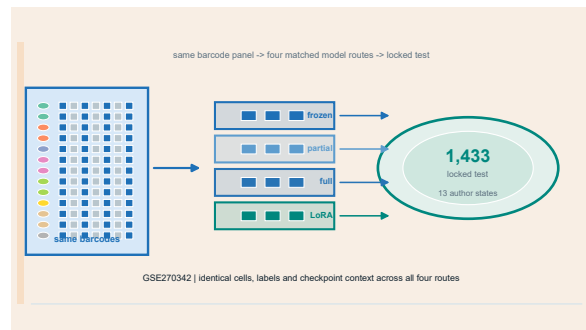
b Sorghum sealed-library contract



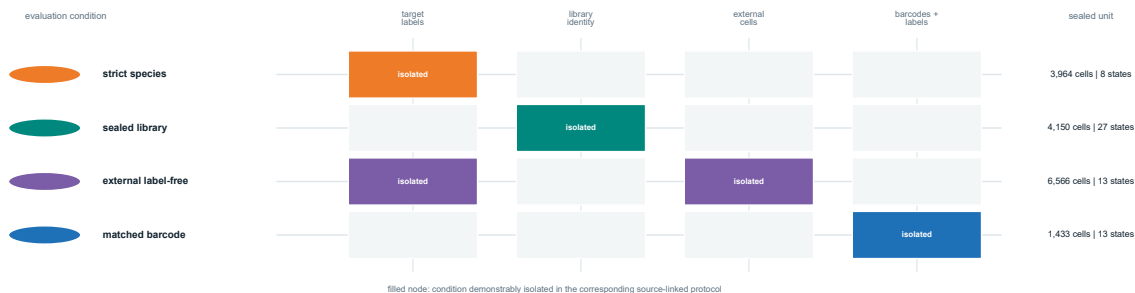
c Independent label-free root inference



d Matched-model barcode control

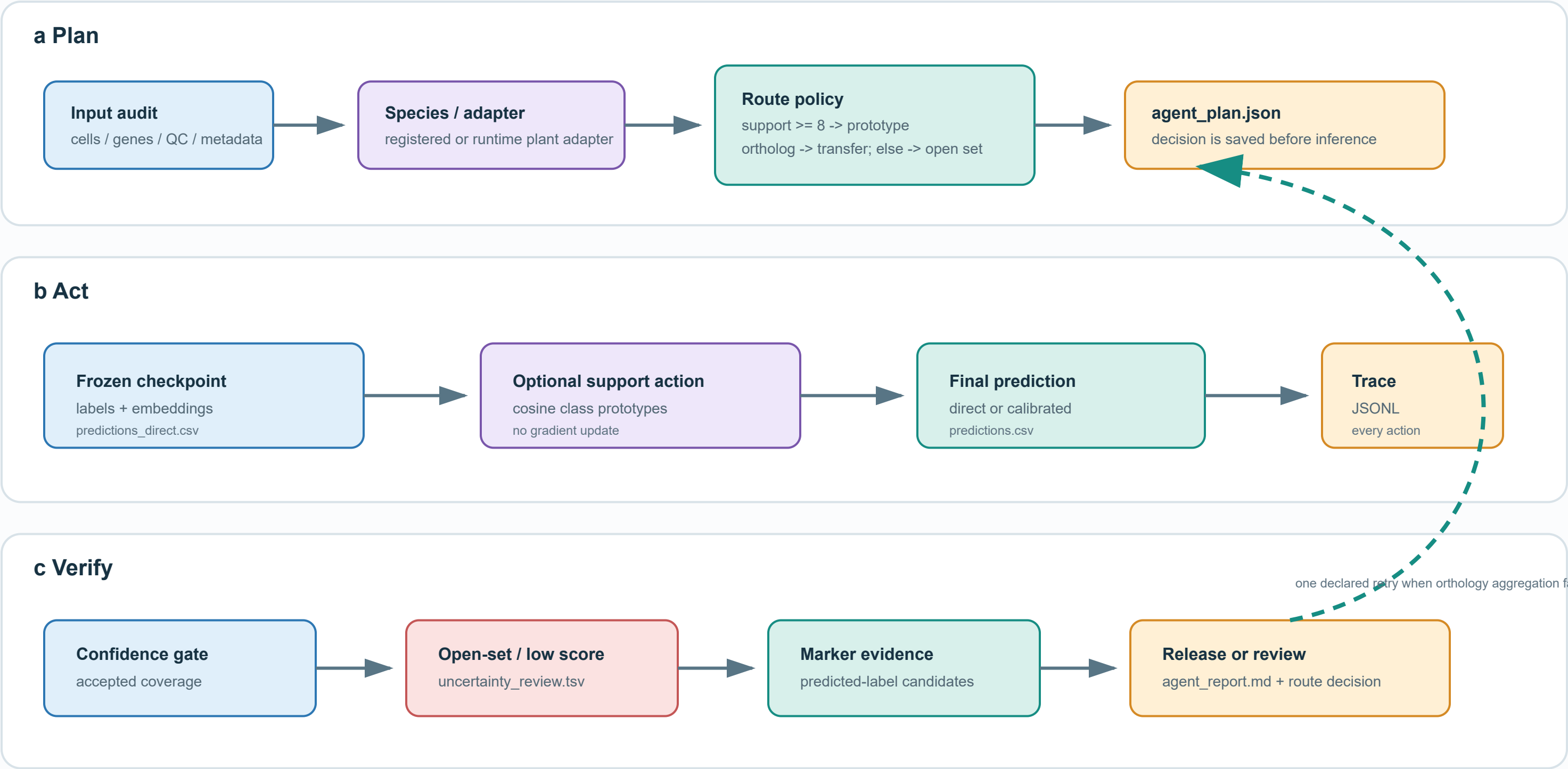


e Independence topology across all evaluations



PlantCell-Agent: evidence-aware annotation loop

Frozen Plant-CellFM inference with explicit routing, selective acceptance and review trace



Direct predictions are retained; selective acceptance is reported separately from all-cell accuracy.